Hospitality Analytics Project

Customer Retention and Dynamic Pricing Analysis in the Hospitality Industry

Prepared By

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Project Duration

4 Weeks

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Power BI
- GitHub

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1. Executive Summary

The hospitality industry faces challenges related to booking cancellations, customer retention, and dynamic pricing. This project analyzes hotel booking data to identify customer behavior patterns, evaluate cancellation trends, optimize pricing strategies, and build a predictive model for booking cancellations.

The analysis was performed using hotel booking data containing customer demographics, booking details, room information, pricing metrics, and reservation outcomes. Data cleaning, exploratory analysis, machine learning, and dashboard development were completed to generate actionable business insights.

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2. Business Problem

Hotel businesses frequently experience:

- High booking cancellation rates
- Revenue loss due to cancellations

- Inefficient pricing strategies
- Difficulty retaining customers
- Seasonal fluctuations in demand

The objective of this project is to identify factors affecting customer cancellations and pricing while providing recommendations for revenue optimization and customer retention.

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3. Project Objectives

The primary objectives are:

1. Clean and preprocess hotel booking data.
2. Analyze customer booking behavior.
3. Identify factors contributing to booking cancellations.
4. Perform customer segmentation analysis.
5. Develop a cancellation prediction model.
6. Create an interactive Power BI dashboard.
7. Provide strategic business recommendations.

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4. Dataset Description

The dataset contains hotel booking information with attributes related to customers, reservations, pricing, and cancellations.

Key Features

- Hotel Type
- Lead Time
- Arrival Date
- Adults
- Children
- Babies
- Country
- Market Segment
- Distribution Channel
- Customer Type
- ADR (Average Daily Rate)
- Booking Changes
- Previous Cancellations
- Reservation Status

Target Variable

- is_canceled

The target variable indicates whether a reservation was canceled.

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5. Data Cleaning and Preprocessing

The following preprocessing steps were performed:

Missing Value Treatment

Missing values in:

- Children
- Country
- Agent
- Company

were handled using appropriate replacement methods.

Duplicate Removal

Duplicate booking records were identified and removed.

Privacy Protection

The following personally identifiable information (PII) columns were removed:

- Name
- Email
- Phone Number
- Credit Card

Feature Engineering

New features were created:

Total Guests

Total Guests = Adults + Children + Babies

Total Stay

Total Stay = Weekend Nights + Week Nights

Total Revenue

Total Revenue = ADR × Total Stay

6. Exploratory Data Analysis

EDA was conducted to understand customer behavior and booking patterns.

Cancellation Analysis

The distribution of canceled and non-canceled bookings was analyzed.

Key observations:

- Customers with longer lead times showed higher cancellation tendencies.
- Certain market segments exhibited significantly higher cancellation rates.

Market Segment Analysis

Bookings were analyzed across:

- Online Travel Agents
- Corporate
- Direct
- Groups

Results showed that some segments contribute disproportionately to cancellations.

Country Analysis

Customer distribution by country was evaluated to identify major source markets.

Monthly Booking Trend

Monthly booking trends revealed seasonal demand fluctuations and peak booking periods.

Pricing Analysis

ADR was analyzed across:

- Months
- Hotel Types
- Market Segments

Pricing variation indicated strong seasonality effects.

7. Machine Learning Model

Objective

Predict whether a hotel booking will be canceled.

Target Variable

is_canceled

Selected Features

- Lead Time
- ADR
- Adults
- Children
- Total Stay
- Previous Cancellations
- Booking Changes

Model Used

Logistic Regression

Data Split

- Training Data: 80%
- Testing Data: 20%

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Results

1. Model Accuracy: 73%
2. Weighted Precision: 69%
3. Weighted Recall: 73%
4. Weighted F1-Score: 66%

The model demonstrated strong capability in identifying cancellation risks.

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8. Power BI Dashboard

An interactive Power BI dashboard was developed consisting of four pages.

Page 1: Executive Overview

KPIs:

- Total Bookings
- Total Revenue
- Average ADR
- Cancellation Rate

Visualizations:

- Monthly Booking Trend
- Hotel Type Distribution

Page 2: Cancellation Analysis

Visualizations:

- Cancellation by Market Segment
- Cancellation by Country
- Lead Time vs Cancellation
- Customer Type Analysis

Page 3: Dynamic Pricing Analysis

Visualizations:

- ADR by Month
- ADR by Market Segment
- Revenue by Hotel Type
- ADR vs Total Guests

Page 4: Customer Segmentation

Visualizations:

- Customer Type Distribution

- Repeat Guests Analysis
- Market Segment Distribution
- Country Analysis

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9. Key Findings

The analysis revealed the following:

1. Long lead times significantly increase cancellation probability.
2. Certain market segments exhibit higher cancellation rates.
3. Repeat guests demonstrate better retention behavior.
4. ADR varies considerably across seasons and customer segments.
5. Seasonal demand patterns strongly influence revenue generation.

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10. Business Recommendations

Based on the analysis:

Cancellation Reduction

- Implement flexible rebooking options.
- Offer incentives for confirmed bookings.
- Introduce predictive cancellation alerts.

Revenue Optimization

- Apply dynamic pricing during peak seasons.
- Increase ADR strategically during high-demand periods.

Customer Retention

- Develop loyalty programs for repeat guests.
- Provide personalized offers based on booking history.

Market Segment Focus

- Prioritize high-value segments with lower cancellation risk.
- Optimize marketing investments for profitable customer groups.

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11. Project Outcome

The project successfully:

- Cleaned and transformed hotel booking data.
- Identified major cancellation drivers.
- Analyzed customer and pricing behavior.
- Built a cancellation prediction model.
- Developed an interactive Power BI dashboard.
- Generated actionable business recommendations.

The insights derived from this analysis can assist hospitality businesses in improving customer retention, reducing cancellations, and maximizing revenue.

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12. Future Enhancements

Future improvements may include:

- Advanced machine learning models
- Real-time booking prediction systems
- Customer lifetime value analysis
- Forecasting future hotel demand
- Automated pricing recommendation engines

References

- Hotel Booking Demand Dataset
- Python Documentation
- Scikit-Learn Documentation
- Power BI Documentation
- Pandas Documentation